

18

Exploratory Factor Analysis

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Introduction

Do you fancy yourself pretty good at playing video games? Some research suggests that individual differences in the ability to play video games is associated positively with intelligence (Baniqued et al., 2015); however, other research suggests that it is not (Unsworth et al., 2015). I would say that much of the ‘brain training movement’ is predicated upon the notion that a

positive association between video game playing ability and cognitive ability exists. Of course, the mere establishment of a positive association between video game playing ability and intelligence would not, on its own, support the notion that getting better at video games will increase one's intelligence. Instead, experimental evidence would be required to support a causal connection between the two variables. Nonetheless, it is arguably interesting to wonder about the nature and strength of the association between video game playing ability and intelligence, considering how popular video game playing is in society.

Before determining the nature and strength of the association between video game playing ability and intelligence, it would be important to understand the nature of intelligence, with respect to the number and nature of the dimensions of intelligence. A statistical analysis that can help determine the number and nature of the dimensions underlying a construct is often called factor analysis. Unfortunately, in practice, many people use the term factor analysis, but what they really mean is data reduction analysis. Strictly speaking, factor analysis is only one type of data reduction analysis, in the same way that 'a Kleenex' is just one type/brand of tissue. Consequently, to avoid confusion, ideally, people would use the term 'data reduction analysis' to refer to the large category of analyses used to reduce the number of observed variables in a dataset to a smaller number of underlying dimensions (or statistical variables). An observed variable is a variable that has scores that you can see in your data file. For example, reaction time, participant responses to survey questions, and annual earnings are all observed variables. By contrast, an underlying dimension is not observable directly in a data file. Instead, it is inferred to exist on the basis of the inter-correlations between observed variables. That's the purpose of all data reduction analyses, including factor analysis: to transform the observed variable correlation matrix into a smaller number of underlying dimensions (statistical variables, if you will). In some cases, the data reduction analysis will uncover one underlying dimension; in others, it will uncover several underlying dimensions.

In addition to factor analysis, people often use component analysis as a data reduction technique. Factor analysis and component analysis are similar, and they often yield similar results. However, there is an important technical difference between these two types of data reduction analyses. In this chapter, I describe and illustrate several different types of factor analysis, as well as one type of component analysis.

I note, here, that I discuss in this chapter data reduction methods that are 'exploratory' in nature, rather than 'confirmatory'. An exploratory data reduction analysis implies that the researcher does not have a strong sense of how many dimensions underlie data. By contrast, a confirmatory data reduction analysis would imply that a fairly strong idea about the number and nature of the dimensions can be hypothesized and tested statistically. There is at least one other important distinction between exploratory and confirmatory factor analysis, and it pertains to the fact that a typical exploratory factor analysis is unrestricted (i.e., all observed variables are free to load onto all dimensions), whereas a confirmatory factor analysis has some restrictions

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imposed by the researcher (not all observed variables are free to load onto all dimensions). For that reason, exploratory factor analysis is sometimes referred to as an unrestricted factor analysis and confirmatory factor analysis as restricted factor analysis. Confirmatory factor analysis can be conducted in specialist software (i.e., not SPSS), which I may treat in a separate chapter at some point.

Foreshadowing the Results

Because there are many steps and terms associated with a dimension reduction analysis, I think it is beneficial to take a peek at the key final results associated with the intelligence test dimension reduction analysis. As can be seen in the SPSS table entitled 'Pattern Matrix', there are nine intelligence subtests listed, from Letter-Number Sequencing to Figure Weights. Furthermore, there are three columns of results. The results consist of coefficients, usually called factor loadings. Factor loadings are similar to standardized beta-weights from multiple regression. In this example, the first column of coefficients is defined by relatively large values from Letter-Number Sequencing (.930), Digit Span Backward (.587), and Digit Span Sequencing (.557). All three of these tests are memory tests. Consequently, I could name the first column of results a memory dimension. The second column is defined by relatively large coefficients from Coding (.903), Symbol Search (.720) and Cancellation (.400) – all tests of processing speed. Hence, I could name the second column of results a processing speed dimension. Finally, the third column is defined by relatively large coefficients from the Visual Puzzles (.907), Matrix Reasoning (.646), and Figure Weights (.551) – all tests of non-verbal reasoning. Thus, I could name the third column of results a non-verbal reasoning dimension. Thus, the data reduction analysis results consist of three dimensions of intelligence: memory span, processing speed, and non-verbal reasoning.

Pattern Matrix^a

	Factor		
	1	2	3
Letter-Number Sequencing	.930	-.110	-.006
Digit Span Backward	.587	.160	.038
Digit Span Sequencing	.557	.129	.178
Coding	.030	.903	-.089
Symbol Search	.102	.720	-.004
Cancellation	-.058	.400	.084
Visual Puzzles	-.069	-.043	.907
Matrix Reasoning	.095	.101	.646
Figure Weights	.201	.036	.551

Extraction Method: Maximum Likelihood.

Rotation Method: Oblimin with Kaiser Normalization.

a. Rotation converged in 7 iterations.

The second key portion of the results pertains to the correlations between the dimensions. As can be seen in the SPSS table entitled, 'Factor Correlation Matrix', all three dimensions of cognitive ability inter-correlated positively. Thus, higher scores on memory span

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(dimension 1) were associated ($r = .440$) with higher scores on processing speed (dimension 2). Additionally, higher scores on memory span (dimension 1) were associated ($r = .587$) with higher scores on non-verbal reasoning (dimension 3). Finally, higher scores on processing speed (dimension 2) were associated ($r = .538$) with higher scores on non-verbal reasoning (dimension 3).

Factor Correlation Matrix

Factor	1	2	3
1	1.000	.440	.587
2	.440	1.000	.538
3	.587	.538	1.000

Extraction Method: Maximum Likelihood.

Rotation Method: Oblimin with Kaiser Normalization.

Thus, based on a data reduction analysis of the nine intelligence subtests included in this fictitious study, three dimensions of intelligence were identified, each defined by three subtests. Furthermore, all three dimensions inter-correlated positively. The above results are the core of a dimension reduction analysis. Although perhaps straightforward, there are many decisions made and steps executed to obtain these core results. I cover all of the steps in this chapter.

Exploratory Factor Analysis: Steps

In practice, conducting an exploratory data reduction analysis involves the following eight steps: (1) determine whether the data are appropriate for a data reduction analysis; (2) decide the number of dimensions to extract from the data; (3) decide the estimation method to generate the solution; (4) decide how to rotate the solution (if at all); (5) interpret the dimensions by examining the most salient loadings; (6) examine the correlations between the dimensions (depending on the rotation method); (7) consider the possibility of conducting a second-order data reduction analysis on the correlations between the dimensions; (8) take a well-deserved break. In this chapter, I will describe how to conduct all eight of these steps.

Appropriate for Data Reduction?

The fundamental calculations of a data reduction analysis are based on the inter-correlations between observed variables. In the context of this example, the nine intelligence subtests are the observed variable. If there are no inter-correlations between the observed variables, it is pointless to conduct a data reduction analysis. Furthermore, theoretically, if one or more underlying dimensions are identified in a data reduction analysis, it *may* be suggested that the underlying dimensions caused the observed variable inter-correlations. Consequently, it is important to evaluate whether the correlation matrix associated with a collection of observed variables is appropriate for data reduction. Stated alternatively, the issue of data reduction appropriateness pertains to whether the observed variables of interest are

correlated sufficiently to merit conducting a data reduction analysis. This is an important consideration, as applying a dimension reduction analysis, such as a factor analysis, to a correlation matrix made-up of a collection of correlations that are essentially zero (within sampling fluctuations) can suggest, erroneously, the presence of a seemingly meaningful dimension. Don't fall into this trap! There are two commonly employed approaches to the evaluation of the dimension reduction appropriateness of a correlation matrix: (1) purely statistical, and (2) effect size.

Statistical Approaches to Data Reduction Appropriateness

A commonly used purely statistical approach to the evaluation of the factorability of a correlation matrix is Bartlett's test of sphericity. Bartlett's test of sphericity tests the hypothesis that a correlation matrix is an identity matrix, i.e., a correlation matrix of zeros – within sampling fluctuations. It is important to emphasize 'within sampling fluctuations', as it is highly unlikely that one will obtain a correlation matrix made-up entirely of observed correlations equal to precisely .00. In a sense, Bartlett's test of sphericity may be regarded as an omnibus test – it tests the statistical significance of all of the correlations in a correlation matrix, simultaneously.

A serious limitation associated with Bartlett's test of sphericity is that it is influenced heavily by sample size. Thus, a correlation matrix with a trivial amount of shared variance between two or more of the observed variables may be found to be statistically significantly different from an identity matrix. As data reduction analyses are typically conducted on data derived from large sample sizes (say, $N > 300$), the issue of "excessive power" is a real one. Consequently, there is good reason to consider alternative approaches that are not influenced by sample size.

Effect Size Approaches to Data Reduction Appropriateness

A commonly used effect size approach to the evaluation of the appropriateness of a correlation matrix for data reduction is the Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy. The KMO index represents the sum of the squared partial correlations associated with all of the observed variable combinations in the correlation matrix. In plain language, the larger the difference between the observed correlations and the partial correlations, the more likely there is one or more dimensions underlying the data. Theoretically, the KMO measure of sampling adequacy can range from .00 to 1.0, however, in practice, .50 should be considered the null hypothesis. A KMO value of .50 would imply that the average correlation between the observed variables is .00. Consequently, Kaiser and Rice (1974) provided the following guidelines to interpret the KMO index: between .00 to .49 = unacceptable; .50 = miserable; .60 = mediocre; .70 = middling; .80s = meritorious; .90 = marvelous. On this basis, I recommend that the KMO index reach at least $\approx .65$ for the purposes of defending an observed correlation matrix as appropriate for data reduction.

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Most programs also provide a measure of sampling adequacy for each observed variable in the correlation matrix. In practice, you might find one or two observed variables with very low levels of sampling adequacy (say, $< .65$). You can consider omitting those observed variables from the analysis. Then, re-run the KMO index analysis to see what you get. Perhaps it will now be sufficiently high to merit a data reduction analysis.

Appropriate for Data Reduction? SPSS

As data reduction analysis is based on the correlations between the observed variables, it is best to first examine the correlation matrix. Consequently, for this example, I ran an analysis to obtain the correlation matrix with all nine intelligence subtests (**Data File: [wais_video_games](#)**). As I was interested only in intelligence, at this stage, I excluded the video games variable from the analyses. I will come back to video game playing ability later in the chapter.

As can be seen in the SPSS table entitled 'Correlation Matrix', all of the correlations between the intelligence subtests were positive. Furthermore, most were fairly substantial in size ($> .30$). Based on a sample size of 200, a correlation greater than $|.13|$ is significant statistically. Clearly, many of the correlations were significant statistically. Thus, I expect Bartlett's test of sphericity to suggest that the correlation matrix to be appropriate for data reduction.

Correlation Matrix									
	Matrix Reasoning	Figure Weights	Visual Puzzles	Letter-Number Sequencing	Digit Span Backward	Digit Span Sequencing	Symbol Search	Cancellation	Coding
Matrix Reasoning	1.00	.55	.63	.42	.44	.46	.40	.16	.40
Figure Weights	.55	1.00	.56	.46	.36	.49	.36	.24	.33
Visual Puzzles	.63	.56	1.00	.36	.36	.46	.34	.23	.31
Letter-Number Sequencing	.42	.46	.36	1.00	.58	.61	.30	.11	.25
Digit Span Backward	.44	.36	.36	.58	1.00	.53	.39	.17	.37
Digit Span Sequencing	.46	.49	.46	.61	.53	1.00	.40	.19	.40
Symbol Search	.40	.36	.34	.30	.39	.40	1.00	.33	.66
Cancellation	.16	.24	.23	.11	.17	.19	.33	1.00	.36
Coding	.40	.33	.31	.25	.37	.40	.66	.36	1.00

In fact, as can be seen in the SPSS table entitled 'KMO and Bartlett's Test', Bartlett's test of sphericity was significant statistically, $\chi^2(36) = 691.860$, $p < .001$. Thus, the correlation matrix may be considered appropriate for data reduction. Furthermore, the Keiser-Meyer-Olkin Measure of Sampling Adequacy was estimated at .847, which is greater than the minimum value of .65 I recommended. Thus, again, the correlation matrix appears to be clearly appropriate for data reduction.

KMO and Bartlett's Test		
Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.847
Bartlett's Test of Sphericity	Approx. Chi-Square	691.860
	df	36
	Sig.	.000

Number of Dimensions

Once the correlation matrix is concluded to merit a data reduction analysis, one must decide the number of dimensions to extract from the correlation matrix. Recall that the purpose of the data reduction analysis is to reduce the number of observed variables in the correlation matrix to the smallest number of underlying (statistical) dimensions that can account for the shared variance between the observed variables (Gorsuch, 1983). In this example, there are nine observed variables in the correlation matrix. Thus, the number of dimensions that one might want to extract from the correlation matrix should be a fair bit less than nine. In the context of an exploratory factor analysis, the number of dimensions to extract from the correlation matrix is determined empirically, based on specialized tests to help researchers with this issue. A large number of tests have been proposed over the years to identify the number of dimensions to extract from the correlation matrix. In this chapter, I will cover the mostly commonly used tests: (1) Kaiser-Guttman rule; (2) scree plot; (3) parallel analysis.

Kaiser-Guttman Rule

The Kaiser-Guttman rule is also known as the eigenvalue greater than 1.0 rule. To understand the logic of the Kaiser-Guttman rule, it is necessary to understand the basic nature of an **eigenvalue**. In any data reduction analysis, there are as many eigenvalues as there are observed variables in the analysis. Correspondingly, the total amount of variance that can be accounted for in a data reduction analysis is equal the number of observed variables included in the data reduction analysis. Thus, in this example with the nine cognitive ability subtests, the total amount of variance in the analysis is equal to 9.0. An eigenvalue represents the amount of variance accounted for by a dimension, or factor/component, in a data reduction analysis. Larger eigenvalues indicate dimensions that account for relatively large amounts of variance. The first dimension extracted from the data reduction analysis will be associated with the largest eigenvalue. Kaiser-Guttman's rule suggests that the number of dimensions to extract from the data reduction analysis is equal to the number of eigenvalues that have a value greater than 1.0, because that is the number of dimensions that accounted for more variance than a single observed variable. Stated alternatively, an eigenvalue less than 1.0 would imply that the dimension has accounted for less variance than one observed variable – not worth extracting.

Theoretically, Kaiser-Guttman's rule makes sense. Furthermore, many statistical software programs use the Kaiser-Guttman's rule as the default. However, comprehensive simulation research has found the Kaiser-Guttman's rule tends to overestimate the number of dimensions to extract from the data (Zwick & Velicer, 1986; Velicer, Eaton & Fava, 2000). Consequently, it is not recommended for use in exploratory factor analysis (Fabrigar, Wegener, MacCallum, & Strahan, 1999). For thoroughness, I have reported the eigenvalues associated

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with the nine intelligence variables included in the analysis. As can be seen in the SPSS table entitled 'Total Variance Explained', the 'Initial Eigenvalues' ranged in size from 4.179 to .303. Furthermore, only two of the eigenvalues exceeded 1.0, which would suggest the extraction of two dimensions from the correlation matrix. Again, I would not trust the Kaiser-Guttman rule.

Total Variance Explained

Component	Initial Eigenvalues			Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	4.179	46.436	46.436	4.179	46.436	46.436
2	1.218	13.531	59.967	1.218	13.531	59.967
3	.911	10.127	70.094	.911	10.127	70.094
4	.732	8.130	78.223			
5	.515	5.723	83.947			
6	.429	4.764	88.711			
7	.359	3.993	92.704			
8	.353	3.927	96.630			
9	.303	3.370	100.000			

Extraction Method: Principal Component Analysis.

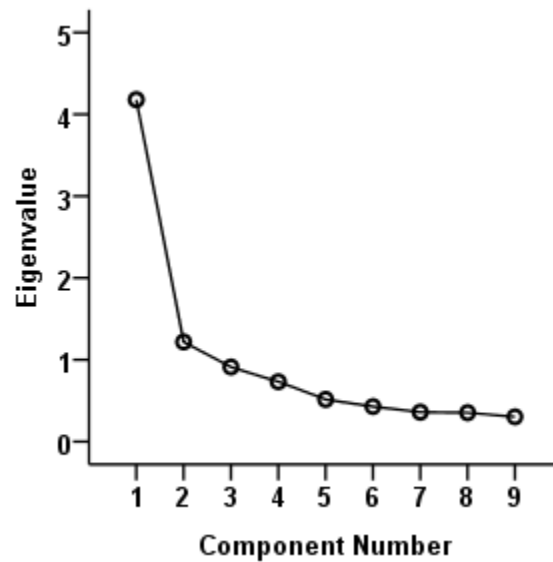
Scree Plot

Cattell (1966), the inventor of the scree plot, contented that the purpose of factor analysis is to determine the number of dimensions that can account for “non-trivial common variance” (p. 245). Thus, Cattell (1966) was responding to the tendency for the K-G rule to lead to the extraction of trivial dimensions. Consequently, Cattell (1966) devised a method that would provide researchers with a justification to leave behind the “rubble” of dimensions that are not worth examining.

A scree plot consists of the rank ordered eigenvalues on the X-axis and the magnitude of the eigenvalue on the Y-axis. A correlation matrix appropriate for reduction will yield a scree plot with a break, or elbow, in the rank ordered eigenvalues. The eigenvalues on the left-side of the break in the ordered eigenvalues are recommended for extraction from the correlation matrix. By contrast, the eigenvalues to the right of the break are considered “rubble”, a mixture of unique and non-shared error variance – not worth extracting.

As can be seen in Figure C18.1 ([Watch V18.1: Scree Plot](#)), the scree plot suggested at least one clear break in the eigenvalues, between eigenvalue 1 and 2 (or ‘Component Number’ 1 and 2). However, there might also be a break between eigenvalue 4 and 5. That’s the problem with scree plots: all too often, it comes down to a bit of an art, rather than a science. In fact, the inter-rater reliability of recommendations using the scree plot technique may be as low as .60 (Zwick & Velicer, 1986). For this reason, the scree plot has been superseded by the parallel analysis.

Figure C18.1. Scree Plot



Parallel Analysis

The beauty of a parallel analysis is that it combines the intuitive appeal of a scree plot with the rigors of statistical hypothesis testing. The key to conducting a parallel analysis is to generate rank ordered eigenvalues derived from randomly generated data. Then, you identify the point at which the raw data (i.e., the data you collected) eigenvalues dip down across the randomly generated eigenvalues. All of the raw data eigenvalues above the randomly generated eigenvalues are considered to be significant statistically and should be extracted for the data reduction analysis.

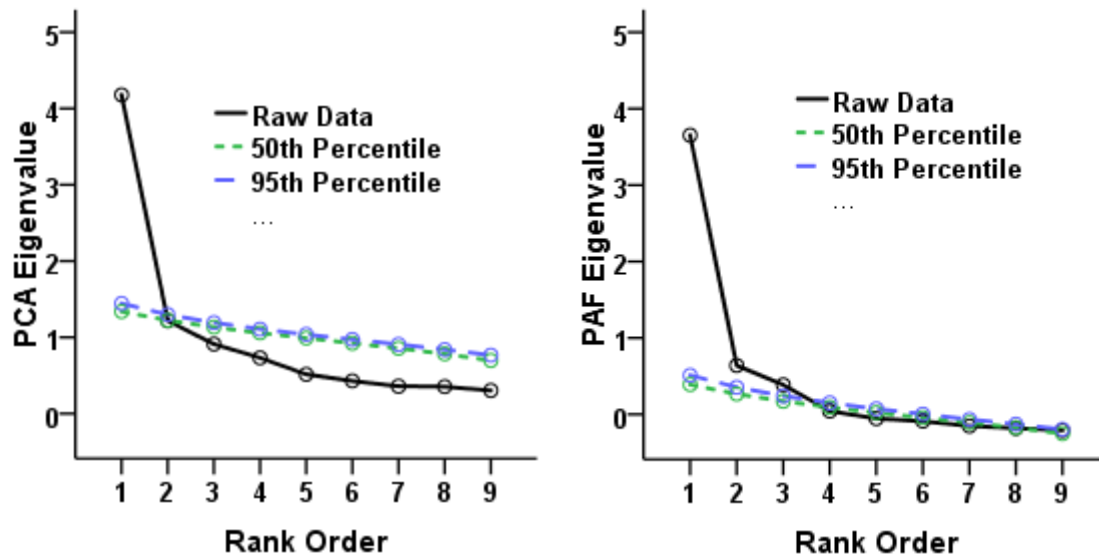
When conducting a parallel analysis, the main thing a researcher has to decide is what level of confidence they want to test the statistical significance of the eigenvalues. In practice, this means either comparing the raw data eigenvalues against the 50th percentile or the 95th percentile of randomly generated eigenvalues. Given the historical popularity of $\alpha = .05$ in inferential statistics, it would be consistent to use the 95th percentile randomly generated data eigenvalues for comparisons. However, historically, parallel analyses have been conducted with the 50th percentile. Perhaps surprisingly, simulation research suggests that the 95th percentile is the best option, when evaluating the first eigenvalue/dimension. Subsequent eigenvalues/dimensions are best evaluated with the 50th percentile, especially if the dimensions are inter-correlated fairly substantially (Crawford et al., 2010). Finally, Crawford et al. (2010) found that parallel analysis eigenvalues derived from principal axis factor estimation was superior to eigenvalues derived from principal component estimation. I discuss estimation methods in more detail in the next subsection.

In order to carry-out a parallel analysis, it is usually necessary to use a separate, dedicated program. For example, O'Connor (2000) created a SPSS syntax file that can be used to conduct a parallel analysis, which includes several options ([Watch V18.2: Parallel Analysis](#)). The options include: (1) number of re-samples (recommendation: 1000); (2) desired percentile comparison (recommendation: 50th percentile, if you think there might be two or more inter-correlated dimensions); (3) estimation method, principal component analysis versus principal axis factor analysis (recommendation: principal axis factor analysis); (4) normally distributed random data versus permutations of the raw data (recommendation: permutations of the raw data, if the observed scores are distributed fairly non-normally). To conduct a parallel analysis on the intelligence data, I applied O'Connor's (2000) program twice for educational purposes. Once with principal components estimation and once with principal axis factor analysis estimation. Recall that Crawford et al.'s (2010) work suggests that principal axis factor analysis estimation should yield more valid results.

As can be seen in Figure C18.2 (Panel A), only one of the raw data principal component estimation eigenvalues was above the random data generated eigenvalues. Consequently, such results suggest that one dimension should be extracted from the data reduction analysis. By contrast, as can be seen in Panel B, the principal axis factor analysis parallel analysis suggested that there were three dimensions to extract from the data reduction analysis, as three of the raw data eigenvalues were above the random data generated eigenvalues. As mentioned above, Crawford et al.'s (2010) work suggests that the principal axis factor analysis estimation approach can be trusted to identify more accurately the number of dimensions to extract. Furthermore, I know from previous research that the nine cognitive ability tests included in the analysis should measure three dimensions. Consequently, I decided to extract three dimensions from the data reduction analysis.

Figure C18.2. Parallel Analysis

Panel A: Component Estimation	Panel B: Factor Estimation
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Estimation Methods

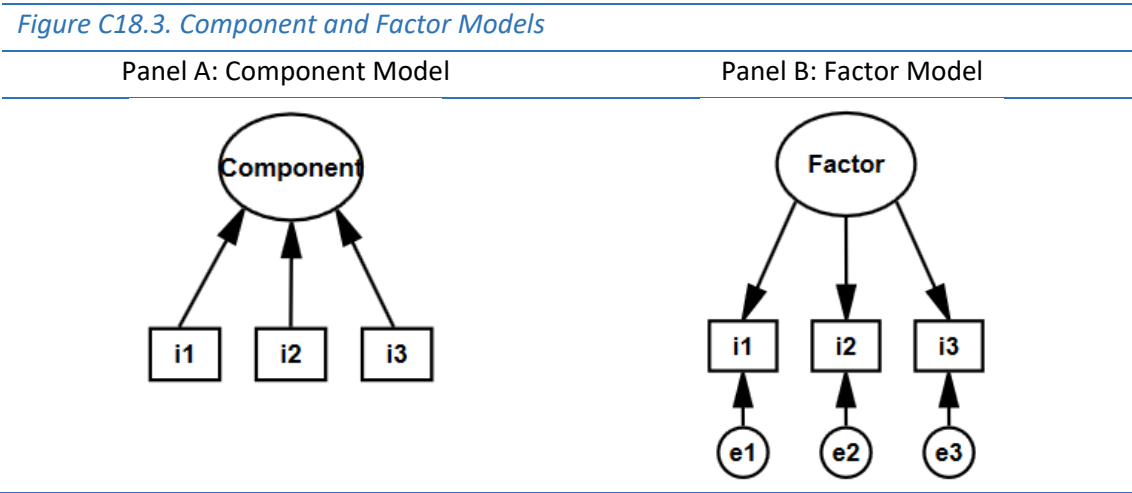
By necessity, I had to mention estimation methods in the previous section, before describing them. That is, when discussing the parallel analysis procedure, I had to mention principal components estimation and principal axis factor estimation, because O'Connor's (2000) program gives the option to use either of them. Importantly, it makes a difference which estimation method one uses.

Estimation methods in data reduction fall into two categories: (1) component analysis; and (2) factor analysis. Within each category, there are several other specific types of estimation methods. Component analysis estimation methods deal with the total variance of variables. Consequently, when looking at a correlation matrix, each variable is associated with variance = 1.0 (you'll see those values on the main diagonal of the correlation matrix). Furthermore, the total variance in the data is calculated on that basis. Thus, in the context of component analyses, people are determining the percentage of the total variance that was accounted for by the number of dimensions (components) extracted from the correlation matrix. Superficially, it may seem like a good thing that component analysis is based on the total variance of variables. In fact, there are advantages associated with component analysis. For example, component analysis does not suffer from the problem of [factor indeterminacy](#). Additionally, a component analysis does not suffer from the possibility of a [Heywood case](#) (proportion of variance accounted for in an observed variable that exceeds 100% - which is impossible).

Arguably, the main problem with component analysis is that it does not really provide the answer to the question researchers typically pose, in the context of data reduction. That is, the data reduction analysis is based on correlations between variables. Thus, it is based on *common* variance, not total variance. In practice, the amount of common variance shared by

observed variables is much less than 1.0 (i.e., less than 100% of the total variance). Therefore, it seems unreasonable to suppose that a data reduction analysis will ever account for 100% of the total variance in the observed scores. Instead, it seems more plausible and consistent to conduct the analysis at the common variance level. Such an approach to data reduction estimation is known as a common factor analysis.

Another important difference between component analysis and factor analysis is that factor analysis explicitly models and partitions common variance and error variance. Error variance, in this context, does not necessarily imply random variance. Typically, the error variance associated with each variable in the model is a mixture of unique reliable variable and random error variance. As can be seen in Figure C18.3 (Panel A), the component model has three observed variables (i1, i2, i3) and one dimension (Component). Furthermore, the lines lead from the observed variables to the component. By contrast, the factor model has three observed variables (i1, i2, i3) and three error terms (e1, e2, e3). Furthermore, the lines lead from the factor to the observed variables, implying that the associations between the observed variables is caused by the factor.



In practice, conducting a common factor analysis implies replacing the 1.0 on the main diagonal of a correlation matrix (i.e., each variable's total standardized variance = 1.0) to another value less than 1.0. Thus, in practice, when a data reduction analysis replaces the 1.0s on the main diagonal of a correlation matrix with a value less than 1.0, it is known as common factor analysis, or just factor analysis. Consequently, component analysis and common factor analysis really are different in that respect. However, in practice, the two techniques will often yield similar data reduction results. However, there are some occasions where they will yield somewhat different results.¹ Naturally, some hard core statisticians do get uptight about the

¹ Based on Widaman (1993), in practice, I'd say the biggest difference between component analysis and factor analysis will be observed when the amount of shared variance between the observed variables is

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difference between component analysis and common factor analysis; so, it's important to be clear about which data reduction estimation method you used.

To the question, to what should we replace the 1.0's in a correlation matrix, in order to conduct a common factor analysis? Good question; one for which there is no definitive answer. Most common factor analysis methods use R^2 : the percentage of variance accounted for by regressing an observed variable onto the remaining observed variables included in the data reduction analysis. The two most popular common factor estimation methods use R^2 : [principal axis factor analysis](#) and [maximum likelihood estimation](#). Based on simulation research, principal axis factor analysis has been found to be superior for solutions that have equal factor loadings, no cross-loadings, and no factor correlations (De Winter & Dodou, 2012). Realistically, you will unlikely find yourself analyzing such data in practice, which would give the nod to maximum likelihood estimation in the majority of cases. The exception is if you are dealing with weak factors. In such cases, principal axis factor analysis appears to identify and represent weak factors better than maximum likelihood estimation (De Winter & Dodou, 2012). In practice, this might mean that you should prefer principal factor analysis, if you are analyzing observed variables that consist of item responses from a questionnaire. In such cases, the factors will tend to be weaker than analyses based on subtests as observed variables.

In the majority of exploratory factor analysis cases, you will probably get along well in your research life knowing principal component analysis, principal axis factor analysis, and maximum likelihood factor analysis. As a general statement, I don't believe there is a substantial benefit to knowing much about the many other estimation methods that have been developed over the years. However, there can be exceptions in specialist cases. All things considered, I recommend that maximum likelihood estimation be used as the default estimation procedure, followed by principal axis factor analysis, if there is a good reason to use it (e.g., it may produce a more theoretically interpretable solution). Lastly, principal component analysis may be considered as a third option, especially if you wish to avoid the factor indeterminacy problem with respect to the estimation of factor scores.

For the purposes of comparison, I conducted the data reduction analysis on the nine intelligence subtests three times ([Watch 18.3: Exploratory Factor Analysis](#)). Once with principal components estimation, once with principal axis factor analysis estimation, and once with maximum likelihood estimation. The corresponding component/factor matrices are reported in Figure C18.4. It can be seen that the component/[factor matrix](#) solutions were very similar across all three estimation techniques. For example, the Matrix Reasoning component loading

relatively small (say, average inter-correlations less than .30). In such scenarios, component analysis will overestimate the magnitude of the loadings and underestimate the magnitude of the correlations between the dimensions. By contrast, common factor analysis will estimate more accurately sized loadings and inter-dimension correlations.

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associated with the first dimension was .75, which was very comparable to the Matrix Reasoning factor loading of .71 obtained from both principal axis factoring estimation and maximum likelihood estimation. Based on the solutions below, all three estimation methods suggested that the first dimension was a general component/factor, as all nine of the intelligence subtests loaded positively on the first dimension. The second dimension was a processing speed dimension, and the third dimension was possibly a memory span dimension, depending on which solution you place more faith. Ultimately, the factor solutions were fairly messy. For example, with respect to maximum likelihood estimation, the Letter-Number Sequencing subtest loaded positively on the first dimension (.70), negatively on the second dimension (-.45) and negatively on the third dimension (-.30). The fact that Letter-Number Sequencing evidenced appreciable loadings across all three dimensions makes for a difficult solution to interpret. Ultimately, the factor matrix solutions that the data reduction analysis first produces is not necessarily the most interpretable. Consequently, statisticians have developed methods to rotate a component/factor matrix solution to achieve greater interpretability. I cover this topic next.

Figure C18.4. Component/Factor Solutions Associated with Three Estimation Methods

Principal Component Analysis^a

	Component		
	1	2	3
Matrix Reasoning	.75	-.15	.34
Figure Weights	.73	-.16	.35
Visual Puzzles	.71	-.15	.51
Letter-Number Sequencing	.69	-.41	-.34
Digit Span Backward	.70	-.19	-.42
Digit Span Sequencing	.76	-.22	-.22
Symbol Search	.68	.48	-.17
Cancellation	.40	.62	.14
Coding	.66	.54	-.17

Extraction Method: Principal Component Analysis.^a

a. 3 components extracted.

Principal Axis Factor Analysis^a

	Factor		
	1	2	3
Matrix Reasoning	.71	-.09	-.26
Figure Weights	.67	-.13	-.20
Visual Puzzles	.69	-.14	-.46
Letter-Number Sequencing	.68	-.41	.36
Digit Span Backward	.65	-.12	.24
Digit Span Sequencing	.72	-.16	.16
Symbol Search	.64	.43	.09
Cancellation	.33	.27	-.04
Coding	.64	.56	.11

Extraction Method: Principal Axis Factoring.^a

a. 3 factors extracted. 21 iterations required.

Maximum Likelihood^a

	Factor		
	1	2	3
Matrix Reasoning	.71	-.03	.29
Figure Weights	.67	-.11	.22
Visual Puzzles	.68	-.08	.50
Letter-Number Sequencing	.70	-.45	-.30
Digit Span Backward	.66	-.14	-.19
Digit Span Sequencing	.73	-.16	-.11
Symbol Search	.64	.41	-.12
Cancellation	.33	.27	.02
Coding	.64	.56	-.16

Extraction Method: Maximum Likelihood.^a

a. 3 factors extracted. 5 iterations required.

Rotation Methods

Once you have decided on the estimation method, which will probably be maximum likelihood, you need to consider how to rotate the factor matrix solution. The reason you need to consider how to rotate the factor matrix solution is because the factor matrix solution provided to you by the data reduction analysis will unlikely be the most interpretable. Thus, applying a rotation technique is about enhancing the interpretability of the results. A rotated factor matrix solution is known as a pattern matrix solution.

There are many different types of rotation methods in the area of data reduction. However, most have the same exact goal: to increase the degree of [simple structure](#) in the solution. Simple structure refers to the degree to which a variable in the data reduction analysis has one especially salient loading on one dimension and relatively small loadings on the remaining dimensions. Thus, ideally, each variable in a data reduction analysis will define substantially only one dimension. In practice, however, one or more variables will associate with a couple of dimensions, in a non-negligible way, which makes interpreting the original solution more difficult. The application of a rotation method can, in many cases, solve the problem by increasing simple structure. Rotation won't always solve the problem, however. In some cases, one or more variables relate so substantially with two or more dimensions that it can't be rotated away.

There are two broad types of rotation methods: orthogonal and oblique. Orthogonal rotation methods produce rotated factor matrix solutions with dimensions that are uncorrelated. In my view, the only reason orthogonal rotation methods were generated, many years ago, was because they were computationally simpler to carry-out (computers were not widely available). A very popular method of orthogonal rotation is varimax rotation (Kaiser, 1958). However, even the inventor of varimax rotation has abandoned it in favor of the development of oblique rotation methods (Harris & Kaiser, 1964).

An oblique rotation method allows dimensions to be correlated with each other. In the behavioral sciences, this is an important allowance, because many dimensions are expected to inter-relate with other. It is important to note that oblique rotation methods will not *force* dimensions to be inter-related. They will only allow the dimensions to inter-relate, if it helps achieve simple structure. Thus, oblique rotation methods are the best of both worlds. If an uncorrelated (orthogonal) solution is best, oblique rotation methods will produce uncorrelated dimensions. However, if a correlated solution best achieves simple structure, oblique rotation methods will produce correlated dimensions. So, why not only ever use an oblique rotation method? Arguably, you should!²

There are two popular widely recommended methods of oblique rotation: direct oblimin (Jennrich & Sampson, 1966) and promax (Hendrickson & White, 1964). As a note, the

² There is one exception to this recommendation: the bifactor rotation. I discuss the bifactor rotation in the Advanced Topics section of this chapter.

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matrix of coefficients produced by rotating a factor matrix via oblique rotation is known as a **pattern matrix**. In practice, researchers use either direct oblimin or promax, but rarely ever provide a reason why they used one over the other. I suspect that people use both in their analyses and report the pattern matrix they find the most interpretable. That's what I do.

Most programs require the user to specify a delta value to execute a direct oblimin rotation. Delta is a value that impacts the degree of inter-correlation allowable between the dimensions. The default delta value in SPSS is 0, which will probably serve you well, in most cases. Similarly, users must specify a kappa value, in order to execute a promax rotation. The first calculations involved with conducting a promax rotation is to estimate the varimax rotated solution. Then, the varimax solution is converted into an oblique solution via a process that involves, amongst other things, exponentiation. Kappa represents the degree to which the initial varimax loadings are exponentiated. The default kappa in SPSS is 4, which, again, will probably serve you well in most cases. In my experience, I tend to get somewhat better results with promax rotation than direct oblimin rotation. Others prefer promax because of its relative conceptual and computational simplicity (Lee & Ashton, 2007). As can be seen below, the direct oblimin (delta = 0) pattern matrix yielded three dimensions: memory span, processing speed, and non-verbal reasoning. Correspondingly, the promax (kappa = 4) pattern matrix yielded three dimensions: memory span, non-verbal reasoning, and processing speed. There was some tendency for the promax solution to produce large factor loadings. Perhaps excessively large loadings (e.g., .99).

Direct Oblimin

Pattern Matrix^a

	Factor		
	1	2	3
Letter-Number Sequencing	.93	-.11	-.01
Digit Span Backward	.59	.16	.04
Digit Span Sequencing	.56	.13	.18
Coding	.03	.90	-.09
Symbol Search	.10	.72	.00
Cancellation	-.06	.40	.08
Visual Puzzles	-.07	-.04	.91
Matrix Reasoning	.09	.10	.65
Figure Weights	.20	.04	.55

Extraction Method: Maximum Likelihood.

Rotation Method: Oblimin with Kaiser Normalization.

a. Rotation converged in 7 iterations.

Promax

Pattern Matrix^a

	Factor		
	1	2	3
Letter-Number Sequencing	.99	-.06	-.16
Digit Span Backward	.61	.01	.13
Digit Span Sequencing	.58	.16	.10
Visual Puzzles	-.10	.93	-.05
Matrix Reasoning	.07	.66	.09
Figure Weights	.19	.55	.02
Coding	-.02	-.07	.92
Symbol Search	.06	.01	.73
Cancellation	-.09	.10	.41

Extraction Method: Maximum Likelihood.

Rotation Method: Promax with Kaiser Normalization.

a. Rotation converged in 5 iterations.

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Factor Correlation Matrix

If an oblique rotation was selected, you will get results relevant to the correlations between the factors. As can be seen in the SPSS table entitled 'Factor Correlation Matrix', there were positive correlations between all three factors. For example, with respect to the direct oblimin rotation, Factor 1 (Memory Span) and Factor 2 (Processing Speed) correlated at .65. Thus, higher ability on memory span was associated with higher ability on processing speed. In fact, all three factors were inter-correlated positively, suggesting the presence of a general factor (see Advanced Topics for a method to factor analyze an inter-factor correlation matrix). It will be noted that the correlations were similar in magnitude across both the direct oblimin and promax solutions. However, it is important to keep in mind that the Factor 2 was Processing Speed in the direct oblimin solution, whereas Factor 2 was Non-Verbal Reasoning in the promax solution.

Direct Oblimin				Promax			
Factor Correlation Matrix				Factor Correlation Matrix			
Factor	1	2	3	Factor	1	2	3
1	1.00	.44	.59	1	1.00	.65	.53
2	.44	1.00	.54	2	.65	1.00	.55
3	.59	.54	1.00	3	.53	.55	1.00
Extraction Method: Maximum Likelihood. Rotation Method: Oblimin with Kaiser Normalization.				Extraction Method: Maximum Likelihood. Rotation Method: Promax with Kaiser Normalization.			

It is possible to conduct a data reduction analysis on a correlation matrix of correlations between components/factors. Thus, for example, the promax 'Factor Correlation Matrix' could be submitted to a factor analysis. As you might expect, in this case, such a factor analysis would produce a general factor, with all three dimensions loading positively on the general factor. Components/Factors derived from correlations between dimensions are known as higher-order factors. I demonstrate how to conduct a higher-order factor analysis in the Advanced Topics section of this chapter.

Finally, it should be noted that when only a single dimension is extracted from a data reduction analysis, it is impossible to apply a rotation method. That is, for a single dimension solution, simple structure is necessarily achieved, as it is impossible for a variable to have more than one appreciable loading – there's only one dimension.

Structure Matrix

Oblique rotation methods produce a [structure matrix](#), in addition to a pattern matrix. A structure matrix consists of a series of coefficients that represent the correlations between the observed variables and the dimensions/factors. Recall that the factor matrix coefficients and the pattern matrix coefficients are more akin to standardized regression beta-weights, i.e.,

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they represent unique effects. Sometimes, it is beneficial to examine the simple associations between the observed variables and the dimensions, when the dimensions are correlated. For example, in the pattern matrix, you might see that one observed variable contributes uniquely to only one dimension. However, based on the structure matrix, you might see that the same observed variable is correlated positively (or negatively) with two or more of the dimensions. Overall, I place much more emphasis on the interpretation of the pattern matrix, as it will represent the greatest degree of simple structure. However, I avoid suggesting that a variable is related to only one dimension on the basis of the pattern matrix results. As can be seen in the SPSS table entitled, 'Structure Matrix', the promax-based structure matrix yielded positive correlations between all of the nine of the intelligence subtests and all three of the dimensions. For example, although Letter-Number Sequencing was numerically most substantially related to dimension 1 (memory span), it was also non-negligibly related to dimension 2 (non-verbal reasoning; $r = .50$) and dimension 3 (processing speed; $r = .33$).

Structure Matrix

	Factor		
	1	2	3
Letter-Number Sequencing	.87	.50	.33
Digit Span Sequencing	.73	.59	.49
Digit Span Backward	.69	.48	.46
Visual Puzzles	.48	.84	.41
Matrix Reasoning	.55	.76	.49
Figure Weights	.57	.69	.43
Coding	.42	.42	.87
Symbol Search	.45	.45	.77
Cancellation	.19	.27	.42

Extraction Method: Maximum Likelihood.

Rotation Method: Promax with Kaiser Normalization.

When an orthogonal rotation method is used, the factor loadings and the structure matrix loadings are identical. Consequently, programs do not report structure matrices, when orthogonal rotation is used.

Factor Scores

It is possible to calculate scores that represent the dimensions extracted from a correlation matrix. Such scores are known as component or factor scores, depending on whether one conducted a component analysis or a factor analysis. Component and factor scores are scores that are derived from. There are several uses of factors scores, including the determination of where the cases lie on the latent dimension of interest (e.g., How high does case 15 score on the Processing Speed factor?), as well as using the factor scores for hypothesis testing. For example, you might want to test the difference between two or more

groups on the factors scores, or perhaps use the factor scores to predict variables outside the factor analysis, as I demonstrate below.

Arguably, the simplest, most non-sophisticated factor scores can be calculated by multiplying each cases scores by the variables' factor loadings associated with a particular factor. Thus, variables with high loadings on a factor would carry greater weight in the creation of the factor scores. Additionally, a negative factor loading, amongst a collection of positive factor loadings, would reduce a case's score on the factor, especially if the case achieved a large score on the negatively loaded variable. Such an approach to creating factor scores has been found to be relatively more stable (i.e., replicable) than the sophisticated methods of creating factor scores (Grice & Harris, 1998).

More sophisticated factor score estimation approaches are not based directly upon the factor loadings. Instead, they tend to be based on regression-type methods designed to predict where the case lies on the factor of interest in an optimal manner. Theoretically, regression-based methods of factor score estimation are considered more accurate. Another advantage of regression-based methods of factor score estimation is that they are standardized with a mean of 0 and standard deviation of 1. Several sophisticated methods of factor score estimation have been developed. Perhaps the most popular are Thurstone's (least squares) regression factor scores (Thomson, 1935; Thurstone, 1935), Bartlett's scores (Bartlett, 1937), and Anderson-Rubin scores (Anderson & Rubin, 1956).

Thurstone's regression factor scores have the ideal property of being most closely related to the factors of interest. Additionally, Thurstone's method was designed to be applied at the population level, which suggests that it may be regarded as more generalizable (Bartholomew, Deary, & Lawn, 2009). However, even when an orthogonal rotation method is applied to a multi-factor solution, Thurstone's factor scores from different factors may be correlated with other factors. Additionally, the factor scores from different factors can be observed to be inter-correlated (Grice, 2001), which some may view unattractive property.

Bartlett's factor scores are based on maximum likelihood estimation, which, in one sense, makes them more accurate estimates at the population level (i.e., less biased; Bartholomew et al., 2009). However, to confuse matters, Bartlett's method was focused upon the accurate estimation of a factor score for a given individual in the data file, rather than a factor score estimation method that could be applied for any case in the population (Bartholomew et al., 2009) Thus, in some sense, Thurstone's regression factor scores may be regarded as more precise but also more biased. By comparison, Bartlett's scores are less precise but also less biased (Bartholomew et al., 2009). An advantage of Bartlett's factor scores is that they are ensured to be uncorrelated with other factors from a multi-factor (orthogonal) solution; something that can't be said about Thurstone's factor scores.

The Anderson-Rubin method of factor score estimation was an adaptation to Bartlett's method. The adaptation involved the constraint of orthogonality of the factor scores with

other factor scores derived from any rotated multi-factor solution (even oblique). However, the factor scores may be correlated with other factors from the solution. Arguably, the main disadvantage associated with the Anderson-Rubin method is that the correlation between the factor scores and the corresponding estimated factor is lower, in comparison to the Thurstone and Bartlett methods.

All things considered, I prefer Thurstone's regression factor scores, as they are more precise and most closely related to their corresponding factor. The fact that the Thurstone factor score estimation method may not generalize as well to the population, in comparison to Bartlett's scores, is not much of a concern, as it is a rare scenario where a researcher wishes to re-apply a factor estimation solution to another sample. Furthermore, to overcome the problem of inter-correlated factor scores, researchers can use statistical analyses to estimate unique effects associated with inter-correlated factor scores, such as multiple regression. I demonstrate such an example below.

Factor Scores: SPSS

SPSS offers three factor score estimation methods: Regression (Thurstone), Bartlett, and Anderson-Rubin. Each method involves saving the factor scores as variables to the data file ([Watch V18.4: Factor Scores IQ](#)). Based on the intelligence three-factor model promax rotated solution, I estimated all three types of factor scores. For the purposes of illustration, I then inter-correlated the factor scores. As can be seen in Table 1, the Thurstone factor scores were the most substantially inter-correlated, followed by the Bartlett factor scores. The Anderson-Rubin factor scores were entirely uncorrelated, as expected, despite the fact that an oblique rotation method was used.

Table 1: Inter-Correlations Between Factor Scores from Three Estimation Methods

	Thurstone			Bartlett			A-R		
	1.	2.	3.	1.	2.	3.	1.	2.	3.
1. Memory Span	1.0			1.0			1.0		
2. Reasoning	.73	1.0		.55	1.0		.00	1.0	
3. Processing Speed	.59	.63	1.0	.46	.46	1.0	.00	.00	1.0

Note. Thurstone (Thomson) = Regression; A – R = Anderson-Rubin

Next, I conducted a multiple regression with the Thurston factor scores to predict video game playing ability. The model yielded a statistically significant $R^2 = .128$, $F(3, 196) = 9.56$, $p < .001$. Furthermore, only processing speed was found to be a statistically significant unique contributor to the model: memory span, $b = .46$, $\beta = .04$, $p = .678$, semi-partial $r = .03$; reasoning, $b = -.39$, $\beta = -.04$, $p = .738$, semi-partial $r = -.02$; processing speed, $b = 3.94$, $\beta = .35$, $p < .001$, semi-partial $r = .27$. Thus, as represented by factor scores, only processing speed was

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associated with video game playing ability. It should be emphasized that all three factor score variables were related positively to video game playing ability, as estimated via zero-order correlations: memory span $r = .23$, $p = .001$; reasoning $r = .22$, $p = .002$; processing speed $r = .26$, $p < .001$. Such a pattern of results suggest the possibility of a common process, which could be evaluated further via a communality regression analysis (see Multiple Regression chapter).

Another approach to evaluating the association between an external variable and factors is to add the external variable to the factor analysis ([Watch V18.5: Factor Analysis with External Variable](#)). When I did so with a maximum likelihood factor analysis and promax rotation, I obtained the results reported in the SPSS table entitled 'Pattern Matrix' (see below). It can be seen that video game playing ability was related uniquely only with the processing speed factor ($\lambda = .39$). Such a result is consistent in essence to the multiple regression results reported above. Naturally, given the general factor within the data, the corresponding structure matrix revealed positive correlations between the video game playing ability variable and all three factors (not presented here).

Pattern Matrix^a

	Factor		
	1	2	3
Coding	.90	-.03	-.05
Symbol Search	.74	.05	.01
Cancellation	.43	-.10	.09
video_games	.39	.04	-.04
Letter-Number Sequencing	-.14	.99	-.06
Digit Span Backward	.14	.60	.02
Digit Span Sequencing	.10	.57	.16
Visual Puzzles	-.06	-.10	.94
Matrix Reasoning	.08	.07	.67
Figure Weights	.03	.19	.55

Extraction Method: Maximum Likelihood.

Rotation Method: Promax with Kaiser Normalization.

a. Rotation converged in 5 iterations.

Factor Scores and Reliability

Just like regular sum (composite) scores, in practice, factor scores incorporate some measurement error. Many seasoned researchers appear unaware of this. I believe the confusion consists of thinking of factor scores as some sort of latent variable (latent variables are devoid of measurement error). However, latent variables and factor scores are not the same. Correspondingly, you can estimate the internal consistent reliability of component and factor scores with formulae dedicated for that purpose. Armor (1973) provides a formula known as theta for the component score case. It's a pretty simple formula (it's based on the

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eigenvalue). You can watch me using it here ([video: Estimating Reliability for Component Scores](#)). Gorsuch (1980) provides a formula for the common factor score case.

Sample Size Requirements

Cattell (1978) recommended a minimum sample size of 250. Comrey and Lee (1992) recommended the following more nuanced guidelines: 100 = poor; 200 = fair, 300 = good; 500 = very good; and 1000+ = excellent. Correspondingly, Goldberg and Digman (1994) recommended between 500 and 1,000. Somewhat more complicated guidelines are based on a ratio of cases to indicators. For example, Everitt (1975) recommended a minimum of 10 cases per indicator. Therefore, a factor analysis on 10 items would require a minimum sample size of 100.

However, simple rules of thumb for factor analysis sample size requirements are crude at best and misleading at worst, in the same way that multiple regression rules of thumb for sample size cannot be counted upon to be valid. The reason simple rules of thumb do not work, in this context, is that one needs to take into account the amount of shared variance between the indicators included in the factor analysis, as well as the number of indicators that define each of the factors (MacCallum, Widaman, Zhang, and Hong, 1999).

Based on a simulation, Hirschfeld, von Brachel & Thielsh (2014) found that 86% of personality items' factor loadings stabilize at a sample of 500.

Advanced Topics

Number of Dimensions Problem: Model Comparison Approach

Choosing the number of dimensions to extract from the correlation matrix will have a big impact on the nature of the results obtained from the data reduction analysis. Consequently, it is important to address this issue correctly. However, one should keep in mind that, in practice, there is no “truly correct” number of dimensions to extract from the correlation matrix. In practice, data reduction analyses provide an approximate solution to the issue of data reduction (MacCallum, 2003). Consequently, an alternative approach to determining the number of dimensions to extract from the data reduction is the model comparison approach (Preacher, Zhang, Kim, & Mels, 2013).

In practice, the model comparison approach involves the calculation of model close-fit indices to evaluate various data reduction solutions across the extraction of 1, 2, 3, 4, etc. dimensions from the correlation matrix. Model close-fit indices tend to be based the maximum likelihood chi-square statistic. Most model close-fit indices are range bound between .00 and 1.0, which makes them easier to interpret. Preacher et al. (2013) recommended the Root Mean Square of Approximation (RMSEA) and the Akaike Information Criterion (AIC). The

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RMSEA may be regarded as a better indicator of the model's verisimilitude (i.e., degree to which the model accounts for the shared variance in the *current* data set). Confidence intervals can also be calculated for the RMSEA. The AIC may be regarded as a better indicator of the generalizability of the model (i.e., replicability on *future* data sets).

The RMSEA is considered an absolute close-fit index, because it is based exclusively upon the implied model chi-square value and implied degrees of freedom. The word implied, in this context, refers to the potentially theoretically interesting factor model solution. The formula to calculate the RMSEA is presented below:

$$RMSEA = \sqrt{\frac{\chi^2_{Implied} - df_{Implied}}{(N - 1) * df_{Implied}}} \quad (1)$$

Thus, with respect to the intelligence data used in the Foundations section of this chapter, the single, two-, and three-factor model solutions were associated with the following three maximum likelihood chi-square values:

Single-Factor	Two-Factors	Three-Factors																		
Goodness-of-fit Test	Goodness-of-fit Test	Goodness-of-fit Test																		
<table><tr><th>Chi-Square</th><th>df</th><th>Sig.</th></tr><tr><td>153.143</td><td>27</td><td>.000</td></tr></table>	Chi-Square	df	Sig.	153.143	27	.000	<table><tr><th>Chi-Square</th><th>df</th><th>Sig.</th></tr><tr><td>71.024</td><td>19</td><td>.000</td></tr></table>	Chi-Square	df	Sig.	71.024	19	.000	<table><tr><th>Chi-Square</th><th>df</th><th>Sig.</th></tr><tr><td>9.288</td><td>12</td><td>.678</td></tr></table>	Chi-Square	df	Sig.	9.288	12	.678
Chi-Square	df	Sig.																		
153.143	27	.000																		
Chi-Square	df	Sig.																		
71.024	19	.000																		
Chi-Square	df	Sig.																		
9.288	12	.678																		

Higher-Order Factors

Factors extracted from a factor analysis program, as demonstrated with the intelligence data in the Foundations section of this chapter, are known as first-order factors, because observed variables (item scores or subtest scores) define the factors. When an oblique rotation method is applied to the factor solution, correlations are often observed between the factors. It is possible to conduct a factor analysis on the correlations between factors, which can yield one or more second-order factors. A second-order factor is a factor defined by first-order factors.

With respect to the intelligence example described in the Foundations section of the chapter, all three first-order factors were inter-correlated positively. Based on maximum likelihood and direct oblimin rotation, the following correlations were obtained.

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Factor Correlation Matrix

Factor	1	2	3
1	1.000	.440	.587
2	.440	1.000	.538
3	.587	.538	1.000

Extraction Method: Maximum Likelihood.

Rotation Method: Oblimin with Kaiser Normalization.

In order to conduct a second-order factor analysis, I recommend that you use correlations to three decimal places. In a program like SPSS, you need to input the correlations into a data file. In this example, the correlation matrix looks like this:

	 ROWTYPE_	 VARNAME_	 factor1	 factor2	 factor3
1	N		200.000	200.000	200.000
2	MEAN		.000	.000	.000
3	STDDEV		1.000	1.000	1.000
4	CORR	factor1	1.000	.440	.587
5	CORR	factor2	.440	1.000	.538
6	CORR	factor3	.587	.538	1.000

Please note that the data file has to be set-up exactly as per above (e.g., ROWTYPE_), otherwise you'll likely get an error message when you run the analysis. To download the example data file above, click [here](#). You can get SPSS to run any type of factor analysis on a correlation matrix SPSS data file, i.e., first-order factor analysis or second-order factor analysis, etc. However, you need to run the factor analysis from a syntax file. There is a specific line of syntax you need to include in the factor analysis, in order to tell SPSS that you are using a correlation matrix as data ([Watch V18.6: Second-Order Factor Analysis](#)).

When I ran the factor analysis (maximum likelihood estimation) on the correlation matrix, I obtained the following results. First, the KMO was estimated at .677, which is greater than the minimum expectation of .65 to suggest an appropriate correlation matrix for data reduction. Secondly, the scree plot showed clearly that only one factor should be extracted. One factor is all that could be expected from a correlation matrix with only three variables. Finally, based on the SPSS table entitled 'Factor Matrix', it may be suggested that the nature of the second-order factor was a general factor, with loadings that ranged in magnitude from .635 to .847.

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Factor Matrix^a

	Factor
	1
factor3	.847
factor1	.693
factor2	.635

Extraction Method:
Maximum Likelihood.

a. 1 factors extracted. 4
iterations required.

Based on the first-order factor analysis (reported in the Foundations section of the chapter), factor 3 was the non-verbal reasoning factor. Consequently, it may be suggested that the general intelligence factor was most substantially defined numerically by non-verbal reasoning ($\lambda = .847$). By comparison, factor 2, processing speed, was the numerically smallest contributor to the general factor ($\lambda = .635$). I wrote “numerically” because I have not conducted a statistical analysis to test whether any of the factor loadings were statistically significantly different from each other. There’s no way to do that in SPSS. Overall, the factor accounted for 53.4% of the variance and the communalities ranged from .403 to .718.

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